

SciDK Demo Abstract

NCI Office of Data Sharing — Fourth Annual Data Sharing Symposium

Data to Discovery: Leveraging Cancer Data in the "Real World"

NCI Shady Grove Campus, Rockville, MD | November 5-6, 2026

Science Data Kit (SciDK) is an open-source research data platform developed at MIT's Koch Institute core facility for Data Science and Multimodal Integration (DSMI). SciDK addresses a persistent challenge in multimodal research: high-value data becomes effectively irretrievable not because it is deleted, but because the context needed to interpret it is lost when a researcher leaves. SciDK automates indexing of heterogeneous datasets across local, cloud, and HPC storage into a knowledge graph (labeled property graph database), using an open-source library of interpreters for common file formats and multi-file dataset profiles — with support for custom formats and no migration required. Integration with platforms like NExtSEEK, used at DSMI for sample-level provenance tracking, demonstrates SciDK's ability to unify structured and loosely organized research data in a single queryable and curatable graph. Designed with security and data integrity in mind, SciDK provides intuitive tools to curate, clean, and annotate data in place, supporting data publication, downstream analytics, and integration with common AI platforms via built-in dashboards or API.

[FACILITY_URL] | [GITHUB_URL]

Demo Planning Notes

Core Framing

SciDK as the hub that makes sense of all the fragmented entry points researchers already use. The pitch inverts the "yet another tool to learn" objection — researchers keep using SharePoint, EDAtools, instrument software, and other existing workflows; SciDK is deployed and curated by a data scientist and pulls everything together, opening up new capabilities without adding burden to the research team.

Real-World Data Entry Points to Highlight

- SharePoint (administrative and project data)
- EDAtools (experimental design and analysis)
- Imaging instrument metadata (DICOM and other instrument-native formats)
- Other analysis and pipeline outputs

Demo Story Arc

- **The problem:** Data enters the lab through many different systems; context is fragmented and lost over time
- **The solution:** SciDK ingests and unifies all of it into a single queryable knowledge graph — no migration, no new workflows for researchers
- **The payoff:** A data scientist deploys and maintains it; researchers and PIs gain dashboards, search, and AI-assisted access to their full institutional dataset

2-Minute Video

- Produce a short overview video to accompany the demo table
 - General-purpose asset: useful for the GitHub landing page, facility page, and any future outreach or introductions to SciDK
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Submission Checklist

Abstract & Submission

- ☐ Add facility URL to abstract
- ☐ Add GitHub URL to abstract
- ☐ Submit abstract form before **August 3, 2026** (travel award eligibility deadline)
- ☐ Confirm whether travel award requires a separate application or if early abstract submission is sufficient

GitHub

- ☐ Decide repo ownership (personal vs. facility account, or both with redirect)
- ☐ Make repo public and presentable
- ☐ Rewrite README as an outside-facing landing page (purpose, quick start, links)

Facility Page

- ☐ Set up or update DSMI branch page for SciDK
- ☐ Cross-link facility page and GitHub

2-Minute Video

- ☐ Script and record a 2-min overview video
- ☐ Embed on GitHub landing page and facility page

At the Symposium

- Demo sessions: **November 5, 2:30–4:00 p.m.** and **November 6, 12:00–1:30 p.m. ET**
- Plan to work the room between and after demo sessions

- 2-min recording requirement is for poster presenters only — does not apply to demo table

Abstract submission deadline: Tuesday, September 1, 2026 Travel award eligibility deadline: August 3, 2026